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Dimensionality reduction and sparse representations: Principal Components Analysis and generalizations

Federico Andreis
Department of Policy Analysis and Public Management
Università Commerciale Luigi Bocconi
federico.andreis@unibocconi.it

McKinsey, Milano
3 November 2017

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Principal Component Analysis

The problem of reducing dimensionality

High-dimensional data are becoming increasingly common: telecommunication, health, finance, and genomics are but a few examples of the fields where the data abundance and complexity can lead to some *disorientation*.

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- ▶ finding subgroups
- ▶ building synthetic indicators
- ▶ predict outcomes

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can become very difficult because of data structure and dimensions.

Dimensionality reduction methods seek to solve this problems by providing low-dimensional representations of high-dimensional data. Today we will discuss the **principal components** (PC) approach.

The classic Principal Component Analysis

Principal Component Analysis (PCA) finds a low-dimensional representation of a dataset that contains as much as possible of its variability.

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The idea: each observation lives in a p dimensional space, but not all of these dimensions are equally interesting. PCA seeks a small number of dimensions that are as interesting as possible (measured by the amount of variability along each dimension).

PCA is an unsupervised learning method that has been successfully employed for tasks as diverse as data compression, pattern recognition, classification, and prediction applied to every research field you can think of.

Mathematical details

The PCs of a data matrix $\mathbf{X}$ are a sequence of projections of its columns, mutually uncorrelated and ordered in variance. They can be seen as linear manifolds approximating a set of n points $x_i \in \mathbf{R}^p$.

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We use singular value decomposition (assuming the column means of $\mathbf{X}$ are zero) to obtain

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}'.$$

$\mathbf{Z} = \mathbf{U}\mathbf{D}$ are the PCs, the columns of $\mathbf{V}$ are the **loadings**. The elements $d_1 \geq d_2 \geq \dots \geq d_p$ of the diagonal matrix $\mathbf{D}$ are proportional to the sample variance of the components.

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The success of PCA is mainly due to two properties:

- ▶ PCs sequentially capture the maximum variability among the columns of $\mathbf{X}$, thus guaranteeing minimal information loss
- ▶ PCs are uncorrelated, so we can talk about one principal components without referring to others.

PC: visual representation with $p = 2$

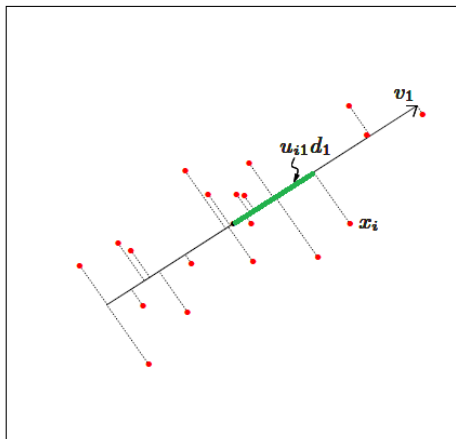


FIGURE 14.20. *The first linear principal component of a set of data. The line minimizes the total squared distance from each point to its orthogonal projection onto the line.*

Example: arrests in the US, 1973 - dataset

Table 1: Sample of the USArrests dataset

	UrbanPop	Murder	Assault	Rape
California	91	9.0	276	40.6
Nevada	81	12.2	252	46.0
West Virginia	39	5.7	81	9.3
Alaska	48	10.0	263	44.5
North Dakota	44	0.8	45	7.3
Maine	51	2.1	83	7.8
Colorado	78	7.9	204	38.7
Virginia	63	8.5	156	20.7
Wyoming	60	6.8	161	15.6
Iowa	57	2.2	56	11.3
South Carolina	48	14.4	279	22.5
New Mexico	70	11.4	285	32.1

Example: arrests in the US, 1973 - PCA

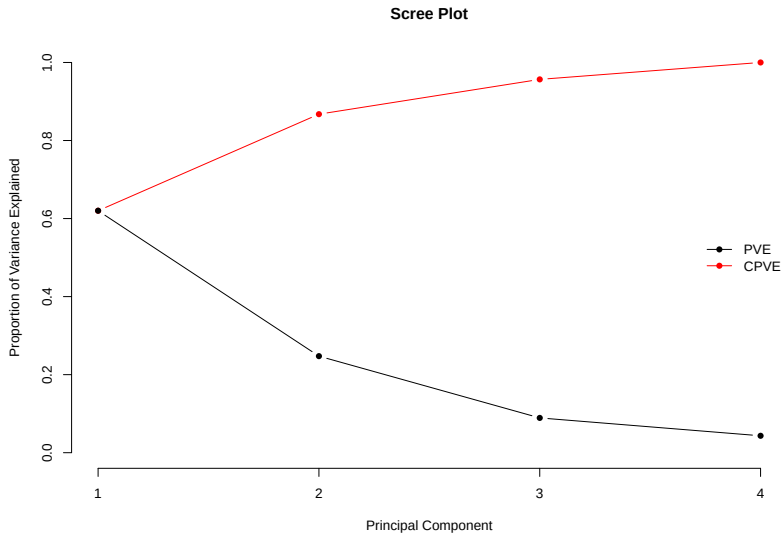
Table 2: PC loadings

	PC1	PC2	PC3	PC4
Murder	0.5359	-0.4182	0.3412	-0.6492
Assault	0.5832	-0.1880	0.2681	0.7434
UrbanPop	0.2782	0.8728	0.3780	-0.1339
Rape	0.5434	0.1673	-0.8178	-0.0890

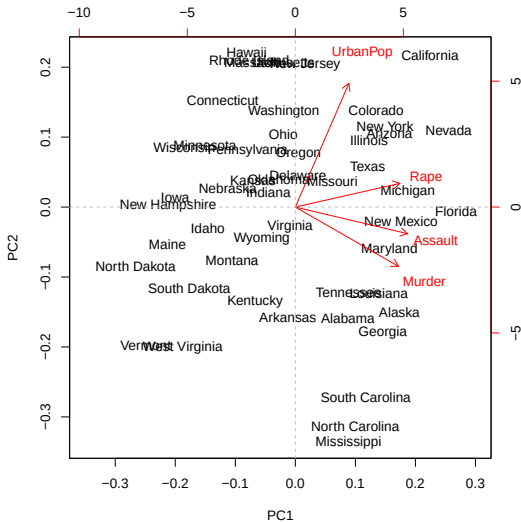
Table 3: Importance of components

	PC1	PC2	PC3	PC4
Standard deviation	1.5749	0.9949	0.5971	0.4164
Proportion of Variance	0.6201	0.2474	0.0891	0.0434
Cumulative Proportion	0.6201	0.8675	0.9566	1.0000

Example: arrests in the US, 1973 - scree plot



Example: arrests in the US, 1973 - biplot



Example: handwritten digits - dataset

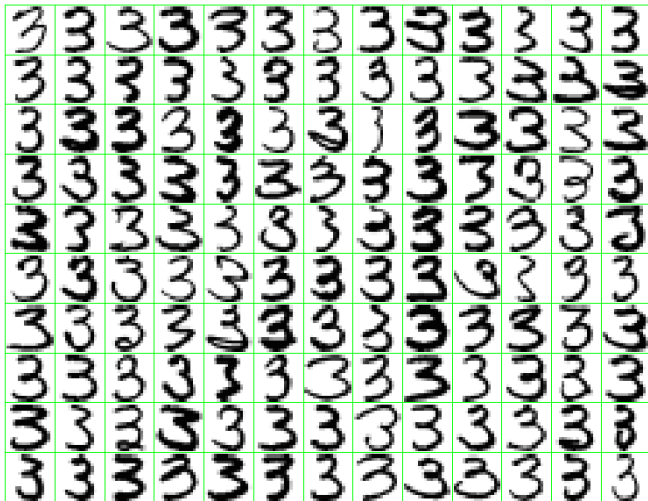


FIGURE 14.22. *A sample of 130 handwritten 3's shows a variety of writing styles.*

Example: handwritten digits - pca results

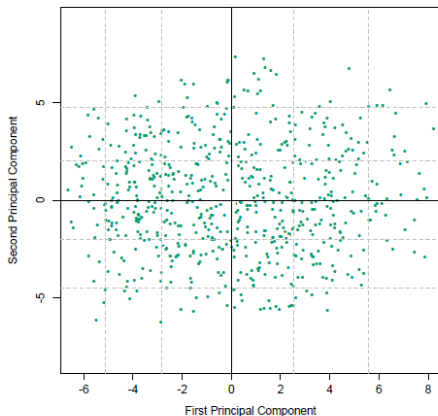


FIGURE 14.23. (Left panel:) the first two principal components of the handwritten threes.

Example: handwritten digits - pca results

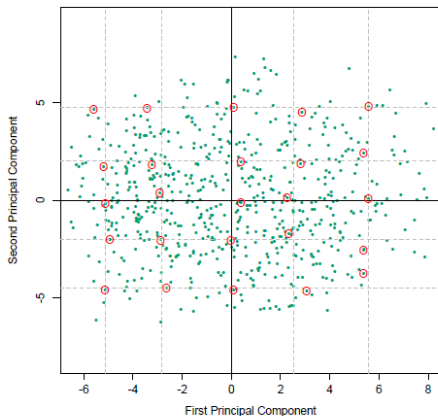


FIGURE 14.23. (Left panel:) the first two principal components of the handwritten threes. The circled points are the closest projected images to the vertices of a grid, defined by the marginal quantiles of the principal components.

Example: handwritten digits - PCA results

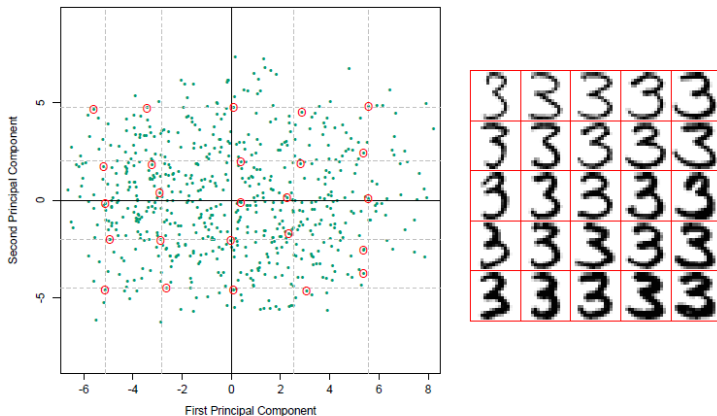


FIGURE 14.23. (Left panel:) the first two principal components of the handwritten threes. The circled points are the closest projected images to the vertices of a grid, defined by the marginal quantiles of the principal components. (Right panel:) The images corresponding to the circled points. These show the nature of the first two principal components.

The issue of interpretability

The last two examples were nice: interpretation of the first two components was intuitive and straightforward. This is, however, not always the case!

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In situations where tens, hundreds, or even thousands of dimensions are at play

- ▶ two components alone are unlikely to capture enough variability
- ▶ interpreting the PCs can be **really** hard.

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- ▶ interpreting the PCs can be **really** hard.

Notwithstanding the nice geometric interpretation, it is not always clear how to fill with meaning these new variables.

Sometimes, strong signals underlie the data, and we get lucky. More often, however, something is needed to mitigate the issue; as a possible solution, we will discuss the **sparse PCA** approach later.

Principal components regression

If we are willing to make the *bet* that along the directions where $\mathbf{X}$ is most variable, so is Y , it may be reasonable to use (some of) the PCs as predictors in a regression model, in lieu of the original data.

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This approach leads to

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- ▶ orthogonal regressors
- ▶ reduced overfitting $\rightarrow$ potentially better test MSE.

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This approach leads to

- ▶ a more parsimonious model
- ▶ orthogonal regressors
- ▶ reduced overfitting $\rightarrow$ potentially better test MSE.

However, there's no free lunch:

- ▶ interpretability issues carry over
- ▶ may be sub-optimal for prediction (PCs are unsupervised).

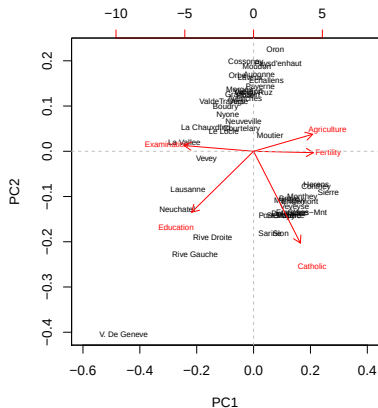
This notwithstanding, **principal components regression** (PCR) and its generalisations are interesting options for prediction.

Example: swiss fertility - dataset

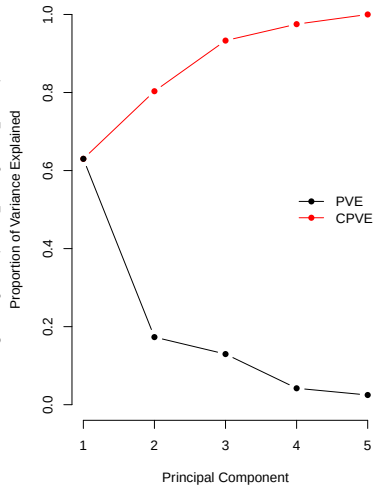
Table 4: Sample of the swiss dataset

	Fert	Agri	Exam	Educ	Cath	Inf_Mor
Courtelay	80.2	17.0	15	12	9.96	22.2
Delemont	83.1	45.1	6	9	84.84	22.2
Franches-Mnt	92.5	39.7	5	5	93.40	20.2
Moutier	85.8	36.5	12	7	33.77	20.3
Neuveville	76.9	43.5	17	15	5.16	20.6
Porrentruy	76.1	35.3	9	7	90.57	26.6
Broye	83.8	70.2	16	7	92.85	23.6
Glane	92.4	67.8	14	8	97.16	24.9
Gruyere	82.4	53.3	12	7	97.67	21.0
Sarine	82.9	45.2	16	13	91.38	24.4
Veveyse	87.1	64.5	14	6	98.61	24.5
Aigle	64.1	62.0	21	12	8.52	16.5

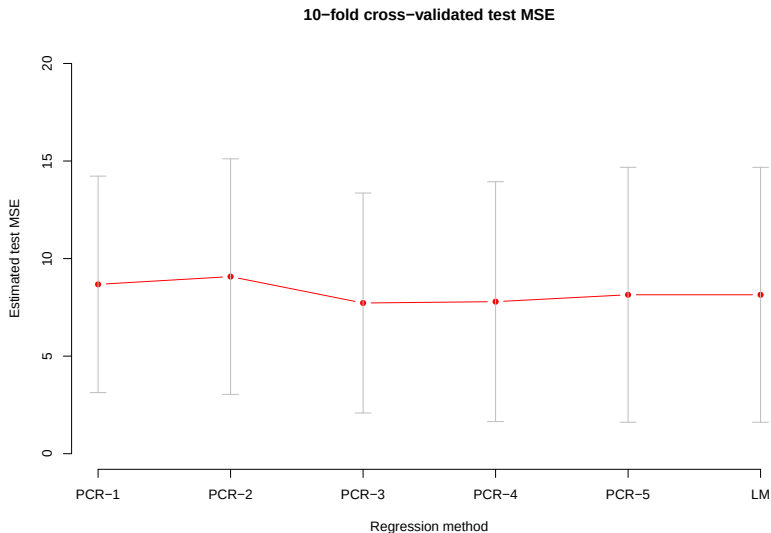
Example: swiss fertility - PCA



Scree Plot



Example: swiss fertility - PCR test MSE



Wrap-up on PCA

Principal component analysis is an unsupervised learning method that seeks to provide low-dimensional informative representations of possibly high-dimensional data. It proved useful for a variety of tasks, examples include:

- ▶ construction of synthetic indicators and data visualisation
- ▶ finding subgroups
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PCs can be difficult to interpret: a deep knowledge of the study subject is of fundamental, yet sometimes not enough. Lastly, classic PCA only deals with numerical variables.

For these reasons, a number of generalisations have been developed; in the following we will discuss a form of **nonlinear PCA** to deal with categorical data and **sparse PCA** to improve interpretability.

Nonlinear principal component analysis

Some drawbacks of PCA

Classic PCA suffers from two important limitations:

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It is hence easy to imagine, that in many real life situations PCA may not always be the most appropriate tool.

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- ▶ it assumes that the relationships between variables are linear
- ▶ it's applicable only if all of the variables are numerical.

It is hence easy to imagine, that in many real life situations PCA may not always be the most appropriate tool.

Alternatives have been developed, that overcome these limitations while still achieving remarkable reductions of the data complexity.

In the following, we present an approach that allows to deal with mixed measurement levels (numerical, ordinal, nominal) that may not be linearly related to each other.

Nonlinear PCA

By **nonlinear principal component analysis** (NLPCA) we here denote an approach to dimensionality reduction in the presence of mixed measurement levels and possibly nonlinear relationships.

The key steps required to perform NLPCA are

1. **choice of analysis level:** define the nature of the variables

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Classic PCA is then applied to the quantified variables.

PCA is a special case of NLPCA, when all variables are treated as numerical and the mapping is linear (standardization).

Analysis levels and quantification

Depending on the analysis levels, different constraints on the quantifications are required

- ▶ **nominal**: same categories → same quantified value
- ▶ **ordinal**: same as nominal, plus respect the ordering
- ▶ **numeric**: same as ordinal, plus respect the spacing.

Analysis levels and quantification

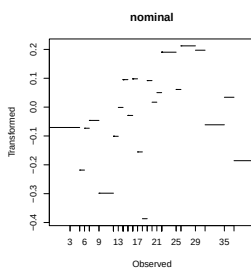
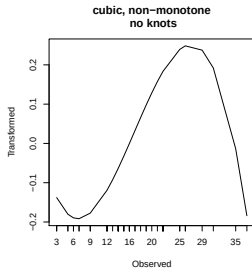
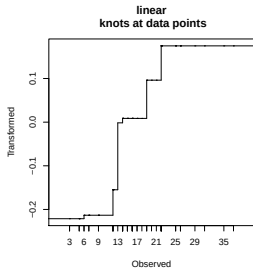
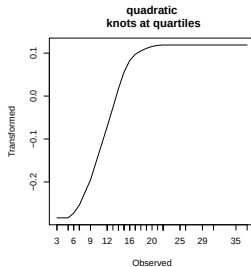
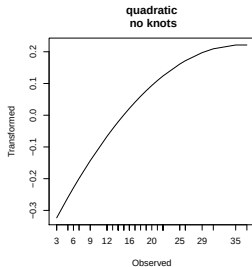
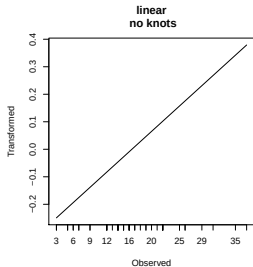
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Optimal quantification determines, sub the above constraints, the values that maximise the first q eigenvalues of the correlation matrix of the quantified variables, where q is a prescribed number of components.

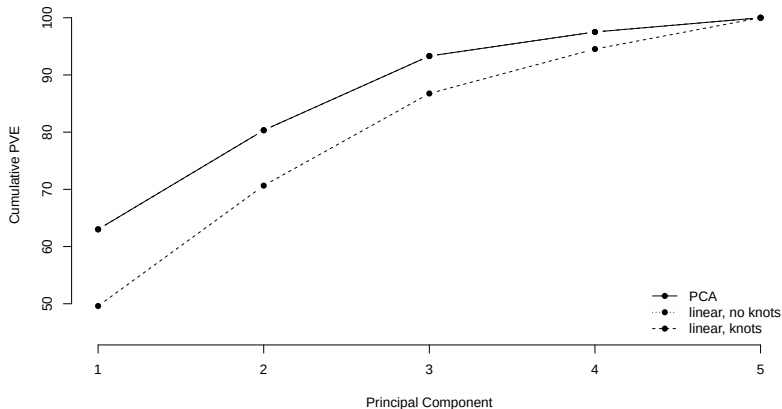
It is also possible to use smooth transformations; splines, that lie between linear and ordinal transformations, are an example.

Transformation plots: swiss dataset - 'Examination'



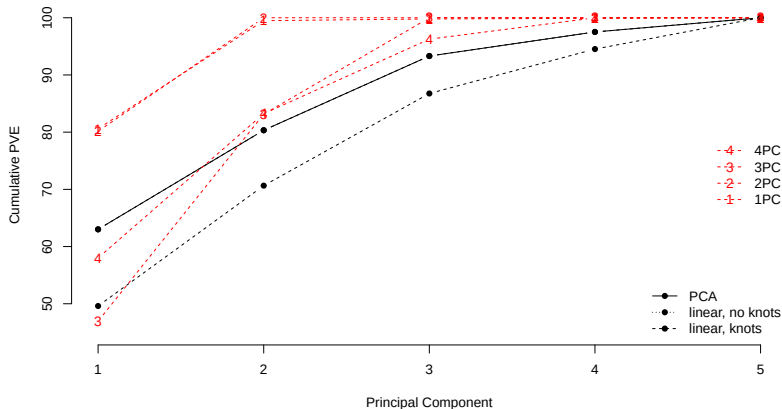
Choosing the number of components

Unlike PCA, different q lead to different results, always do some sensitivity analysis. CV would be great, but computationally too intensive.



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Wrap-up on NLPCA

NLPCA is an unsupervised approach developed as an alternative to classic PCA for handling mixed variables and nonlinearity in measurement scales and relationships.

To use NLPCA one has to choose

- ▶ analysis level (nominal, ordinal, numeric)
- ▶ transformation type.

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- ▶ transformation type.

Moreover, special care must be put in selecting the number of components for optimisation. Although CV could in principle be used to help with these tasks, in practice the computational burden calls for clever solutions.

Note: NLPCA is a names shared by other techniques, such as principal curves and surfaces, kernel pca, functional pca, ...

Sparse Principal Components

On the interpretability of the PCs

Since the components are linear combinations of many variables, and the loadings are typically nonzero, their interpretation can be really difficult.

Researchers often resort to ad-hoc solutions such as rotation techniques to improve readability and arbitrary thresholds on the loadings to reduce the number of explicitly used variables.

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Researchers often resort to ad-hoc solutions such as rotation techniques to improve readability and arbitrary thresholds on the loadings to reduce the number of explicitly used variables.

More general approaches have been developed, based on the fact that PCA can be expressed as a regression-type optimization problem with an ℓ_2 penalty.

Among the various proposals, we here discuss the **sparse PCA**, that uses an elastic net-type penalty (mixed ℓ_1 and ℓ_2) on the component loadings.

Sparse PCA

Let $\mathbf{x}_i$ be the i -th row of the standardized data matrix $\mathbf{X}$. Sparse PCA solves

$$(\mathbf{A}^*, \mathbf{B}^*) = \arg \min_{\mathbf{A}, \mathbf{B}} \sum_{i=1}^n \|\mathbf{x}_i - \mathbf{A}\mathbf{B}'\mathbf{x}_i\|_2^2 + \lambda \sum_{j=1}^p \|\mathbf{b}_j\|_2^2 + \sum_{j=1}^p \lambda_{1,j} \|\mathbf{b}_j\|_1$$

subject to $\mathbf{A}'\mathbf{A} = \mathbf{I}$, where $\mathbf{b}_j$ is the j -th column of $\mathbf{B}$ and is proportional to the j -th component loadings.

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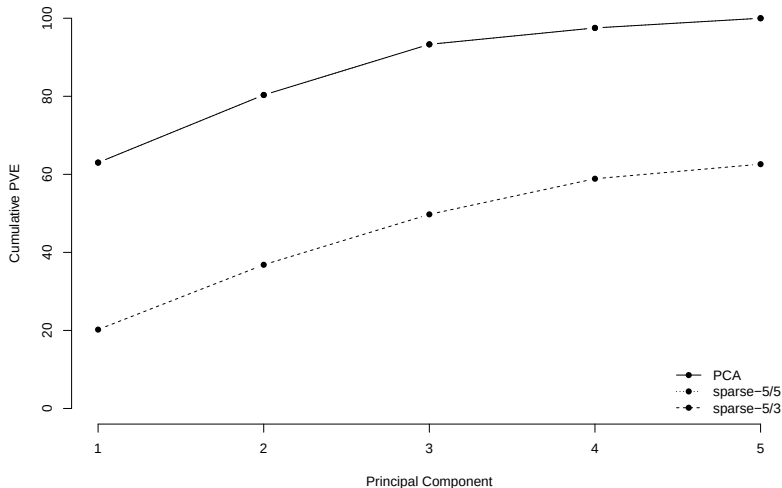
subject to $\mathbf{A}'\mathbf{A} = \mathbf{I}$, where $\mathbf{b}_j$ is the j -th column of $\mathbf{B}$ and is proportional to the j -th component loadings.

In practice, sparse PCA provides an approximation to the classic PCA solution with new components characterised by sparse loadings.

This results in an increased interpretability of the components, at the cost of a loss of explained variance for a given q .

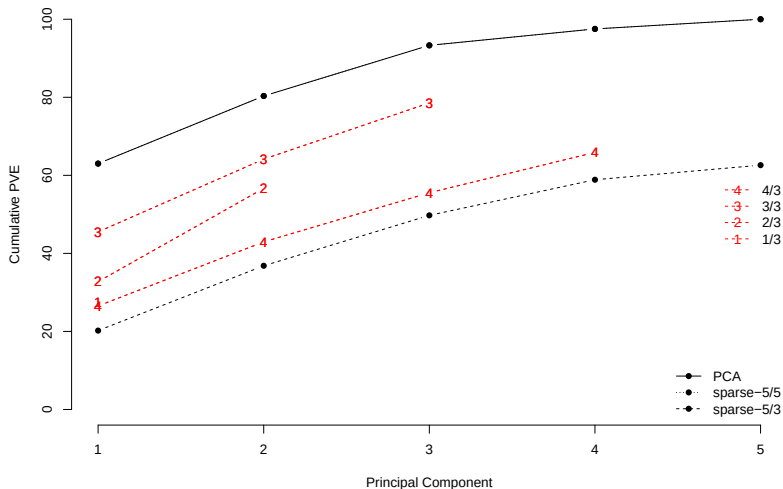
Example: swiss data

Obtaining sparse component loadings comes at a cost in terms of explained variance, as can be readily seen from the scree plots.



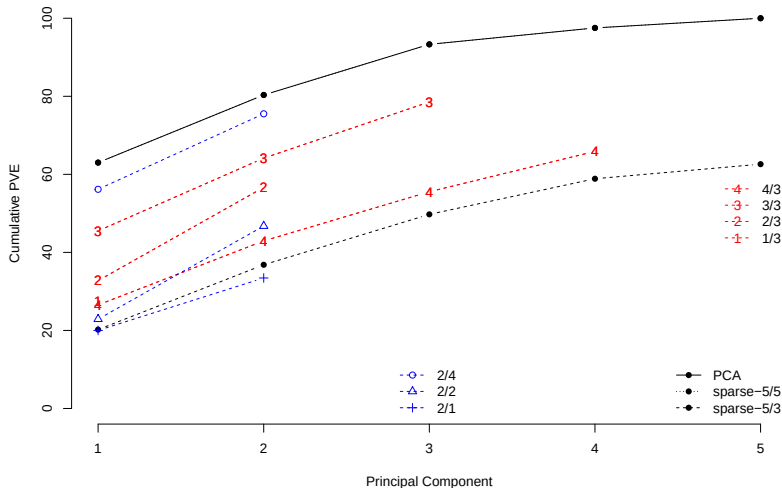
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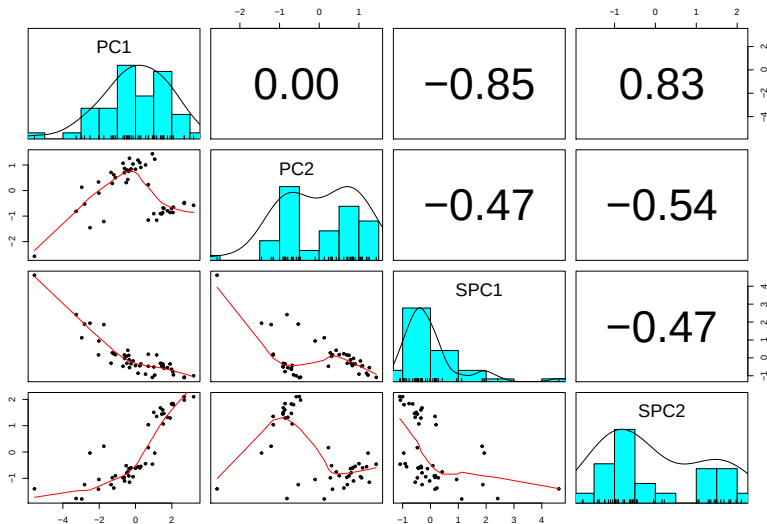


Example: swiss data - loadings

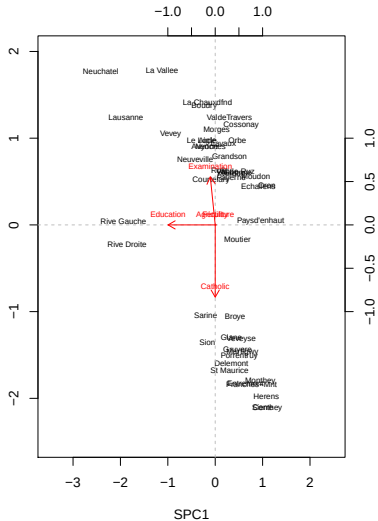
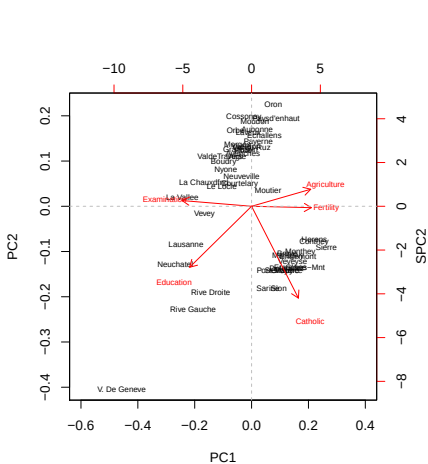
Table 5: Loadings - PCA vs sparse PCA(2/2)

	PC1	PC2	SPC1	SPC2
Fertility	0.4466	-0.0123	0.0000	0.0000
Agriculture	0.4417	0.1525	0.0000	0.0000
Examination	-0.5186	0.0513	0.1040	-0.5538
Education	-0.4627	-0.5457	0.9946	0.0000
Catholic	0.3499	-0.8223	0.0000	0.8327

Example: swiss data - scores



Example: swiss data - biplot



Wrap-up on sparse PCA

Sparse PCA is an extension of classic PCA that encourages, via a lasso penalty, a sparse representation of the components by allowing some of the loadings to be exactly zero.

When performing sparse PCA

- ▶ some parameter tuning is needed
- ▶ some information is lost - sparse PCs approximate PCs.

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When performing sparse PCA

- ▶ some parameter tuning is needed
- ▶ some information is lost - sparse PCs approximate PCs.

However, the use of a sparse representation

- ▶ can improve interpretability of the components
- ▶ yields low-dimensional representations with lower complexity
- ▶ is computationally extremely efficient.

Sparse PCA is especially useful in situations where $p \gg n$ (the analysis of gene data is a typical example).

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